

Setup & Run Guide

Restaurant Table Reservation & Waitlist System

Java Programming - Case Study 70 | Aditya Sunil Chouksey (150096725070)

How to use this guide

Work through the stages in order. Every command is in a two-column table: the left column is exactly what you type, the right column explains what it does.

After each stage there is a Checkpoint telling you what you should see. If you do not see it, stop and look in the Troubleshooting table at the end before continuing.

Stage 6 is your **end-to-end test checklist** - tick each box as you go. It takes about 15 minutes and is a good rehearsal for the live demo.

Stage 0 - What you need

Item	Details
Computer	Windows 10/11, macOS or Linux, with a screen (it is a desktop GUI).
Java Development Kit (JDK)	Version 8 or newer. JDK 17 or 21 is recommended. You need the JDK (it contains javac, the compiler), not only the JRE.
Project folder	"Final Project" containing src/, docs/, run.sh, run.bat and README.md.
Optional	An IDE such as IntelliJ IDEA Community, Eclipse or VS Code with the Java extension.

Your Mac specifically

The `java` command already on your Mac is an old Java 8 runtime without the compiler. Typing `javac` on its own fails with "Unable to locate a Java Runtime that supports javac". A full JDK 21 is installed through Homebrew at `/opt/homebrew/opt/openjdk@21` - Stage 1 shows how to point to it.

Stage 1 - Install or select the JDK

macOS

Type this	What it does
<code>/opt/homebrew/opt/openjdk@21/bin/javac -version</code>	Checks the Homebrew JDK 21 that is already installed on your Mac.
<code>brew install openjdk@21</code>	Only if the line above says "No such file": installs JDK 21 with Homebrew.
<code>export JAVA_HOME=/opt/homebrew/opt/openjdk@21</code>	Tells run.sh (and this terminal window) which JDK to use. Lasts until you close the terminal.
<code>echo 'export JAVA_HOME=/opt/homebrew/opt/openjdk@21' >> ~/.zshrc</code>	Optional: makes JAVA_HOME permanent for every new terminal.
<code>"\$JAVA_HOME/bin/javac" -version</code>	Confirms the compiler that will be used.

Checkpoint

You see a line such as `javac 21.0.x`.

Windows

1. Download "Eclipse Temurin JDK 21 (LTS)" for Windows x64 (.msi) from adoptium.net.
2. Run the installer. On the "Custom Setup" screen enable "Set JAVA_HOME variable" and "Add to PATH".
3. Close and reopen Command Prompt so it picks up the new PATH.

Type this	What it does
<code>javac -version</code>	Checks that the compiler is on the PATH.
<code>java -version</code>	Checks the runtime. Both should report the same major version (for example 21).

Checkpoint

Both commands print version 21 (or whichever JDK you installed). If you see "'javac' is not recognized", see Troubleshooting.

Stage 2 - Open the project folder

Type this	What it does
<code>cd ~/Documents/Java/"Final Project"</code>	macOS: moves the terminal into the project. The quotes are needed because the folder name has a space.
<code>cd "%USERPROFILE%\Documents\Java\Final Project"</code>	Windows Command Prompt equivalent (adjust if you copied the folder elsewhere).
<code>ls</code>	macOS/Linux: lists the folder contents (on Windows use <code>dir</code>).

Checkpoint

You see `README.md docs run.bat run.sh src` (an `out` or `data` folder may also appear after the first run).

Stage 3 - Compile and run with the script

Type this	What it does
<code>./run.sh</code>	macOS/Linux: deletes old compiled classes, compiles everything from <code>src/</code> into <code>out/</code> , and opens the application.
<code>chmod +x run.sh</code>	Only if you get "permission denied": makes the script executable, then run <code>./run.sh</code> again.
<code>run.bat</code>	Windows: does the same as <code>run.sh</code> . (Written for Windows but not yet tried on a Windows PC - if it fails, use Stage 4.)

Checkpoint

A window titled "Restaurant Table Reservation & Waitlist System" opens with four tabs. The Host notifications box at the bottom says "First run: loaded sample data. Changes are saved automatically to ...restaurant.dat".

Within a second a [READY] line appears, for example "Table 5 (6 seats, Main Hall) is free - page waitlisted party W002 - Karan Malhotra".

Stage 4 - Compile and run by hand (what the script does)

Useful if a script fails, and worth understanding for the viva.

Type this	What it does
<code>mkdir out</code>	Creates the output folder for compiled .class files (skip if it already exists).
<code>javac -encoding UTF-8 -d out -sourcepath src src/com/restaurant/Main.java</code>	-encoding UTF-8: read source as UTF-8. -d out: put .class files in out/ (in package folders). -sourcepath src: where to find the other source files; javac compiles every class that Main uses, directly or indirectly.
<code>java -cp out com.restaurant.Main</code>	-cp out: look for classes in out/. com.restaurant.Main: the fully-qualified class whose main() method starts the program.
<code>java -cp out com.restaurant.Main backup.dat</code>	Optional: use a different data file instead of data/restaurant.dat.

Windows note

Use backslashes in the source path: `javac -encoding UTF-8 -d out -sourcepath src src\com\restaurant\Main.java`

Checkpoint

javac prints nothing (or only warnings) and the same window as Stage 3 opens.

Stage 5 - Run inside an IDE (optional)

IntelliJ IDEA

1. File > Open... and choose the "Final Project" folder.
2. Right-click src > Mark Directory as > Sources Root.
3. File > Project Structure > Project: choose your JDK as the SDK.
4. Open src/com/restaurant/Main.java and click the green Run arrow next to main().

VS Code

1. Install the "Extension Pack for Java".
2. File > Open Folder... > "Final Project".
3. Open Main.java and click "Run" above the main() method.

Eclipse

1. File > New > Java Project, untick "Use default location", browse to "Final Project", Finish.

2. If src is not a source folder: right-click it > Build Path > Use as Source Folder.
3. Right-click Main.java > Run As > Java Application.

Checkpoint

The same window opens. The data file is created in the IDE's working directory (normally the project folder).

Stage 6 - End-to-end test checklist

Start from fresh sample data: File > Load Sample Data (or delete the data folder) before TC01. Work top to bottom - some cases build on the previous one. Tick the box when the expected result appears.

Time of day matters

Bookings and walk-ins are only accepted between 11:00 and 22:30, and the sample data adds today's bookings only when it is inside those hours. Run the checklist during that window; tables 1 and 3 may then show today's bookings (Meera Joshi, Arjun Rao).

Done	ID	Do this	You should see
☐	TC01	Floor & Tables > Add Table. Enter number 3, any seats, OK.	Error: "Table 3 already exists."
☐	TC02	Select table 3 > Edit Table > Seats 2 > OK.	Error: "Cannot reduce table 3 to 2 seats: R1001 has a party of 4." (R1001 is Aarav Sharma's booking for tomorrow)
☐	TC03	Reservations: name "Rahul Verma", phone 9811122233, Party size 21, tomorrow 19:00 > Book.	Error: "Party size cannot exceed 20. Larger groups need an event booking with the manager."
☐	TC04	Same form, party 4, Date & time = yesterday's date 19:00 > Book. Then try today 23:00.	"...is in the past." then "Guests are seated between 11:00 and 22:30; 23:00 is outside service hours."
☐	TC05	Same form, Date & time tomorrow 19:07 > Book.	"Reservations are taken in 15-minute slots (e.g. 19:00, 19:15, 19:30)."
☐	TC06	Date & time 2026-02-30 19:00 > Book.	"\"2026-02-30 19:00\" is not a valid date/time. Use yyyy-MM-dd HH:mm..."
☐	TC07	Clear the form. Name "Sara Ali", phone 9822233344, party 3, tomorrow 15:00, Section Any, Table Auto-assign > Book.	Confirmed at a 4-seat table (smallest that fits) - note its table number.
☐	TC08	Name "Dev Rao", phone 9833344455, party 2, tomorrow 15:30, Table = the table from TC07 > Book.	Error: "Table N is already booked by R.... around ... Choose another table or time."
☐	TC09	Walk-ins & Waitlist: name "Ali Khan", phone 9000011111, party 2, Section Patio > Walk-in Arrived.	Patio is full: added to the waitlist with position # and "about N min".

Done	ID	Do this	You should see
☐	TC10	Floor & Tables: select table 8 (Diya Reddy, Patio) > Release Table (guests left). Wait.	"Table 8 released..." then about 2 s later a [READY] line: table 8 held for Isha Verma (W001), banner turns yellow.
☐	TC11	Immediately add another walk-in: party 2, Section Patio > Walk-in Arrived.	Queued behind the others - nobody takes table 8 from Isha.
☐	TC12	Name "Neel Shah", phone 9844455566, party 2, the 15-minute slot 20-30 min from now > Book.	At once its table turns Reserved and a [READY] "Table N is now held for R..." line appears.
☐	TC13	Reservations: type "YA" in Search (Customer name / phone), Sort by Party size.	Only names containing "ya" (Priya, Kavya, Ananya, Diya) - capitals ignored - smallest party first.
☐	TC14	Select Neel Shah's booking from TC12 > Cancel Reservation > Yes.	Status Cancelled; its table is freed and offered to the next suitable party or shown Available.
☐	TC15	Close the window (the app saves). Run ./run.sh again.	Log says "Loaded 12 tables, N reservations and M waiting parties (saved ...)"; everything is as you left it.
☐	TC16	Close the app. In the terminal type: echo broken > data/restaurant.dat - then run the app again.	"Could not read saved data (...). The unreadable file was kept as restaurant.dat.corrupt-... Started with sample data instead."

When everything is ticked: File > Load Sample Data to reset before your demo.

Stage 7 - Data file and resetting

Want to...	Do this
Start again with sample data	File > Load Sample Data (asks for confirmation), or close the app and delete the data folder.
Start with an empty restaurant	File > Clear All Data.
Save immediately	File > Save Now (Cmd+S / Ctrl+S). Normally the app saves 1.5 s after every change and again on exit.
Undo changes since the last save	File > Reload Saved Data.
Find the data file	data/restaurant.dat inside the folder you ran the app from.

Troubleshooting

Problem / message	Likely cause	Fix
"Unable to locate a Java Runtime that supports javac" (macOS)	Only the old Java 8 runtime is on the PATH; no compiler.	export JAVA_HOME=/opt/homebrew/opt/openjdk@21 and run ./run.sh again (Stage 1).

Problem / message	Likely cause	Fix
"javac: command not found" / "'javac' is not recognized"	JDK not installed, or not on PATH.	Install a JDK (Stage 1); on Windows reopen Command Prompt after installing.
UnsupportedClassVersionError	Compiled with a newer JDK but run with an older java (e.g. javac 21 + java 8).	Use java from the same JDK: "\$JAVA_HOME/bin/java" -cp out com.restaurant.Main, or just use run.sh with JAVA_HOME set.
"permission denied: ./run.sh"	Script is not executable.	chmod +x run.sh
"This application needs a graphical display"	Running without a screen (SSH, server, headless container).	Run it on a normal desktop session.
Error: Could not find or load main class com.restaurant.Main	Wrong folder, or out/ not compiled.	cd into the project folder and compile first (Stage 4).
[WARN] Could not save data / "NOT SAVED" in the header	Folder is read-only (e.g. inside a zip or a protected directory).	Extract/copy the project to Documents and run from there.
"Could not read saved data..." on start	data/restaurant.dat is damaged or from an incompatible version.	Nothing to do: the app kept the old file as restaurant.dat.corrupt-... and started with sample data.
Booking refused although tables look free	Outside service hours, not a 15-minute slot, or the table is needed by another booking within 90 min.	Read the error text - it states the rule. Pick another time or table.
Walk-in not seated although a table is green	A reservation needs that table within 90 min, or an earlier waiting party fits it.	Expected behaviour (bookings come first, waitlist is fair).
run.bat fails on Windows	Not yet tried on a Windows PC.	Use the commands in Stage 4 with backslashes, and send me the exact error text.